

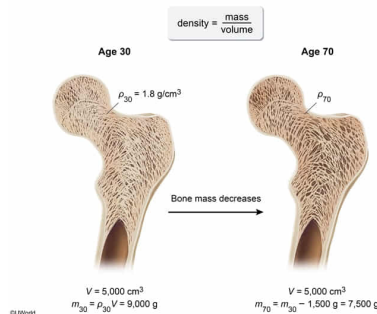
A 30-year-old woman has an average bone tissue density of 1.8 g/cm^3 with an estimated bone tissue volume of $5,000 \text{ cm}^3$. Assuming her bone tissue volume remains the same, what will her bone tissue density be if she is expected to lose 1.5 kg of bone mass by age 70 due to osteoporosis?

- ☐ A. 0.7 g/cm^3 [6%]
☐ B. 1.2 g/cm^3 [14%]
☒ C. 1.5 g/cm^3 [75%]
☐ D. 2.1 g/cm^3 [3%]

Correct

76% answered correctly

Explanation:



The **density** of an object represents the amount of mass contained in a unit of volume, and it has SI units of kg/m^3 . The woman's bone tissue density ρ is calculated as her bone mass m divided by her bone volume V :

$$\rho = \frac{m}{V}$$

The above equation can be rearranged to solve for m . Therefore, her **bone mass** at **age 30** is the product of her bone density at age 30 (1.8 g/cm^3) and her bone volume ($5,000 \text{ cm}^3$):

$$m_{30} = \rho_{30} \cdot V = 1.8 \text{ g/cm}^3 \cdot 5,000 \text{ cm}^3 = 9,000 \text{ g}$$

At age 70, the woman's bone mass is expected to decrease by 1.5 kg. This value is converted to grams using the [unit conversion](#) $1 \text{ kg} = 1,000 \text{ g}$ to determine her new bone mass at **age 70**:

$$m_{70} = m_{30} - 1,500 \text{ g} = 9,000 \text{ g} - 1,500 \text{ g} = 7,500 \text{ g}$$

Assuming the woman's bone volume remains the same, her **bone density** at **age 70** is calculated using $V = 5,000 \text{ cm}^3$ and her new bone mass (7,500 g):

$$\rho_{70} = \frac{m_{70}}{V} = \frac{7,500 \text{ g}}{5,000 \text{ cm}^3} = 1.5 \text{ g/cm}^3$$

(Choice A) Density is calculated as mass divided by volume. If instead the woman's bone volume ($5,000 \text{ cm}^3$) is divided by her bone mass (7,500 g), the result is $0.67 \text{ cm}^3/\text{g}$, which is the *inverse* of density.

(Choice B) The woman's bone volume is given and assumed to be unchanged. If her bone volume is incorrectly calculated as her bone density (1.8 g/cm^3) divided by her expected loss in bone mass (1.5 kg), the result is $1.2 \text{ g}/(\text{kg} \cdot \text{cm}^3)$, which has units of *inverse volume*.

(Choice D) The woman's bone density at age 70 is expected to decrease because her bone volume remains unchanged while her bone mass decreases by 1.5 kg. If her bone mass at age 70 is determined by incorrectly adding 1.5 kg, her bone density is calculated as 2.1 g/cm^3 .

Educational objective:

The density ρ of an object is its mass m divided by its volume V : $\rho = m/V$. This equation can be used to determine an object's mass from density and volume.

Time spent:0 sec
Subject:
Physics

QID:401295
System:
Fluids